University of Delhi

College of Vocational Studies

CYBER FORENSICS

PRACTICAL FILE

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1. Study of Network-Related Commands (Windows)

Command: **ipconfig all**

```
C:\Users\abhishek>ipconfig all

Error: unrecognized or incomplete command line.

USAGE:
    ipconfig [/allcompartments] [/? | /all |
                                 /renew [adapter] | /release [adapter] |
                                 /renew6 [adapter] | /release6 [adapter] |
                                 /flushdns | /displaydns | /registerdns |
                                 /showclassid adapter |
                                 /setclassid adapter [classid] |
                                 /showclassid6 adapter |
                                 /setclassid6 adapter [classid] ]

where
    adapter             Connection name
                       (wildcard characters * and ? allowed, see examples)

    Options:
       /?               Display this help message
       /all             Display full configuration information.
       /release         Release the IPv4 address for the specified adapter.
       /release6        Release the IPv6 address for the specified adapter.
       /renew           Renew the IPv4 address for the specified adapter.
       /renew6          Renew the IPv6 address for the specified adapter.
       /flushdns        Purges the DNS Resolver cache.
       /registerdns     Refreshes all DHCP leases and re-registers DNS names
       /displaydns      Display the contents of the DNS Resolver Cache.
       /showclassid     Displays all the dhcp class IDs allowed for adapter.
       /setclassid      Modifies the dhcp class id.
       /showclassid6    Displays all the IPv6 DHCP class IDs allowed for adapter.
       /setclassid6     Modifies the IPv6 DHCP class id.


The default is to display only the IP address, subnet mask and
default gateway for each adapter bound to TCP/IP.

For Release and Renew, if no adapter name is specified, then the IP address
leases for all adapters bound to TCP/IP will be released or renewed.

For Setclassid and Setclassid6, if no ClassId is specified, then the ClassId is removed.
```

```
Examples:
    > ipconfig                          ... Show information
    > ipconfig /all                     ... Show detailed information
    > ipconfig /renew                   ... renew all adapters
    > ipconfig /renew EL*               ... renew any connection that has its
                                            name starting with EL
    > ipconfig /release *Con*           ... release all matching connections,
                                            eg. "Wired Ethernet Connection 1" or
                                                "Wired Ethernet Connection 2"
    > ipconfig /allcompartments         ... Show information about all
                                            compartments
    > ipconfig /allcompartments /all ... Show detailed information about all
                                            compartments

C:\Users\abhishek>
```

```
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\yash>ipconfig

Windows IP Configuration


Ethernet adapter vEthernet (WSL):

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::131f:8a90:6485:4e87%50
   IPv4 Address. . . . . . . . . . . : 172.29.96.1
   Subnet Mask . . . . . . . . . . . : 255.255.240.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::1f38:a2ff:2c9:fd9c%6
   IPv4 Address. . . . . . . . . . . : 192.168.1.8
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

Command Prompt

```
C:\Users\yash>ping youtube.com

Pinging youtube.com [142.250.192.78] with 32 bytes of data:
Reply from 142.250.192.78: bytes=32 time=28ms TTL=116
Reply from 142.250.192.78: bytes=32 time=28ms TTL=116
Reply from 142.250.192.78: bytes=32 time=29ms TTL=116
Reply from 142.250.192.78: bytes=32 time=27ms TTL=116

Ping statistics for 142.250.192.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 27ms, Maximum = 29ms, Average = 28ms

C:\Users\yash>tracert youtube.com

Tracing route to youtube.com [142.250.192.78]
over a maximum of 30 hops:

  1    <1 ms     9 ms     8 ms  192.168.1.1 [192.168.1.1]
  2     6 ms     2 ms    16 ms  172.26.50.1 [172.26.50.1]
  3     1 ms     2 ms     5 ms  45.116.71.125
  4    23 ms    24 ms    23 ms  103.27.170.10
  5    43 ms    42 ms    40 ms  72.14.239.103
  6    24 ms    24 ms    24 ms  142.250.61.203
  7    27 ms    27 ms    28 ms  bom12s16-in-f14.1e100.net [142.250.192.78]

Trace complete.
```

```
C:\Users\yash>nslookup youtube.com
Server:  192.168.1.1
Address:  192.168.1.1

Non-authoritative answer:
Name:    youtube.com
Addresses:  2404:6800:4009:829::200e
          142.250.192.78
```

Command Prompt

```
C:\Users\yash>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    192.168.1.8:59956      142.250.82.249:3478    ESTABLISHED
  TCP    192.168.1.8:60174      tzdelb-bb-in-f14:https  TIME_WAIT
  TCP    192.168.1.8:60177      40.104.66.210:https    TIME_WAIT
  TCP    192.168.1.8:60178      4.144.165.14:https     ESTABLISHED
  TCP    192.168.1.8:60184      20.44.229.112:https    TIME_WAIT
  TCP    192.168.1.8:60186      20.69.136.49:https     TIME_WAIT
  TCP    192.168.1.8:60187      a23-10-239-251:http    TIME_WAIT
  TCP    192.168.1.8:60188      72.153.5.141:https     TIME_WAIT
  TCP    192.168.1.8:60189      a23-10-239-251:http    TIME_WAIT
  TCP    192.168.1.8:60191      192:domain             TIME_WAIT
  TCP    192.168.1.8:60198      tzdelb-bg-in-f14:https  TIME_WAIT
  TCP    192.168.1.8:60199      192:domain             TIME_WAIT
  TCP    192.168.1.8:60200      192:domain             TIME_WAIT
  TCP    192.168.1.8:60202      192:domain             TIME_WAIT
  TCP    192.168.1.8:60204      192:domain             TIME_WAIT
  TCP    192.168.1.8:60210      192:domain             TIME_WAIT
  TCP    192.168.1.8:60211      nrt12s11-in-f170:https  TIME_WAIT
  TCP    192.168.1.8:60213      192:domain             TIME_WAIT
  TCP    192.168.1.8:60215      192:domain             TIME_WAIT
  TCP    192.168.1.8:60216      192:domain             TIME_WAIT
  TCP    192.168.1.8:60219      192:domain             TIME_WAIT
  TCP    192.168.1.8:60220      192:domain             TIME_WAIT
  TCP    192.168.1.8:60221      192:domain             TIME_WAIT
  TCP    192.168.1.8:60222      192:domain             TIME_WAIT
  TCP    192.168.1.8:60223      192:domain             TIME_WAIT
  TCP    192.168.1.8:60225      tzdela-be-in-f1:https  TIME_WAIT
  TCP    192.168.1.8:60227      tzdelb-bb-in-f14:https  TIME_WAIT
  TCP    192.168.1.8:60228      192:domain             TIME_WAIT
  TCP    192.168.1.8:60229      192:domain             TIME_WAIT
  TCP    192.168.1.8:60230      tzdela-bg-in-f14:https  TIME_WAIT
  TCP    192.168.1.8:60231      192:domain             TIME_WAIT
  TCP    192.168.1.8:60232      192:domain             TIME_WAIT
  TCP    192.168.1.8:60234      192:domain             TIME_WAIT
  TCP    192.168.1.8:60235      tzdela-ax-in-f3:https  TIME_WAIT
  TCP    192.168.1.8:60238      192:domain             TIME_WAIT
  TCP    192.168.1.8:60240      tzdela-bf-in-f14:https  ESTABLISHED
  TCP    192.168.1.8:60241      192:domain             TIME_WAIT
```

Command Prompt

```
C:\Users\yash>netstat -ano

Active Connections

  Proto  Local Address          Foreign Address        State           PID
  TCP    0.0.0.0:135            0.0.0.0:0              LISTENING       452
  TCP    0.0.0.0:445            0.0.0.0:0              LISTENING       4
  TCP    0.0.0.0:5040           0.0.0.0:0              LISTENING       3584
  TCP    0.0.0.0:7680           0.0.0.0:0              LISTENING       6836
  TCP    0.0.0.0:49664          0.0.0.0:0              LISTENING       808
  TCP    0.0.0.0:49665          0.0.0.0:0              LISTENING       660
  TCP    0.0.0.0:49666          0.0.0.0:0              LISTENING       1464
  TCP    0.0.0.0:49667          0.0.0.0:0              LISTENING       1452
  TCP    0.0.0.0:49668          0.0.0.0:0              LISTENING       3232
  TCP    0.0.0.0:49669          0.0.0.0:0              LISTENING       780
  TCP    127.0.0.1:49350        0.0.0.0:0              LISTENING       7024
  TCP    127.0.0.1:49351        0.0.0.0:0              LISTENING       9172
  TCP    172.29.96.1:139        0.0.0.0:0              LISTENING       4
  TCP    192.168.1.8:139        0.0.0.0:0              LISTENING       4
  TCP    192.168.1.8:59956      142.250.82.249:3478    ESTABLISHED     4304
  TCP    192.168.1.8:60302      142.250.183.238:443    TIME_WAIT       0
  TCP    192.168.1.8:60306      142.250.193.74:443     TIME_WAIT       0
  TCP    192.168.1.8:60310      142.250.182.106:443    TIME_WAIT       0
  TCP    192.168.1.8:60317      142.250.193.42:443     TIME_WAIT       0
  TCP    192.168.1.8:60324      142.250.192.227:443    ESTABLISHED     4304
  TCP    192.168.1.8:60333      172.253.118.84:443     TIME_WAIT       0
  TCP    192.168.1.8:60334      142.250.183.238:443    TIME_WAIT       0
  TCP    192.168.1.8:60336      192.168.1.1:53         TIME_WAIT       0
  TCP    192.168.1.8:60337      192.168.1.1:53         TIME_WAIT       0
  TCP    192.168.1.8:60338      142.251.223.129:443    ESTABLISHED     4304
  TCP    192.168.1.8:60339      142.250.193.131:443    TIME_WAIT       0
  TCP    192.168.1.8:60340      192.168.1.1:53         TIME_WAIT       0
  TCP    192.168.1.8:60341      192.168.1.1:53         TIME_WAIT       0
  TCP    192.168.1.8:60342      142.250.192.174:443    ESTABLISHED     4304
  TCP    192.168.1.8:60343      192.168.1.1:53         TIME_WAIT       0
  TCP    192.168.1.8:60344      192.168.1.1:53         TIME_WAIT       0
  TCP    192.168.1.8:60346      192.168.1.1:53         TIME_WAIT       0
  TCP    192.168.1.8:60347      192.168.1.1:53         TIME_WAIT       0
  TCP    192.168.1.8:60541      4.213.25.242:443       ESTABLISHED     3544
  TCP    192.168.1.8:64547      74.125.130.188:5228    ESTABLISHED     4304
  TCP    [::]:135               [::]:0                 LISTENING       452
  TCP    [::]:445               [::]:0                 LISTENING       4
  TCP    [::]:7680              [::]:0                 LISTENING       6836
```

```
C:\Users\yash>arp -a

Interface: 192.168.1.8 --- 0x6
  Internet Address      Physical Address      Type
  192.168.1.1           b8-dd-71-91-cc-6a     dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static

Interface: 172.29.96.1 --- 0x32
  Internet Address      Physical Address      Type
  172.29.104.205        00-15-5d-5e-4c-57     dynamic
  172.29.111.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

C:\Users\yash>route print
===========================================================================
Interface List
 50...00 15 5d cf 87 0d ......Hyper-V Virtual Ethernet Adapter
  8...9e ad 97 c7 60 2d ......Microsoft Wi-Fi Direct Virtual Adapter
 17...9e ad 97 c7 68 2d ......Microsoft Wi-Fi Direct Virtual Adapter #2
  6...6c c2 17 76 f7 51 ......Realtek PCIe FE Family Controller
  5...9c ad 97 c7 60 2d ......Broadcom BCM43142 802.11 bgn Wi-Fi Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.1.1      192.168.1.8     35
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      172.29.96.0    255.255.240.0         On-link       172.29.96.1    271
      172.29.96.1  255.255.255.255         On-link       172.29.96.1    271
  172.29.111.255  255.255.255.255         On-link       172.29.96.1    271
```

Command Prompt

```
C:\Users\yash>route print
===========================================================================
Interface List
 50...00 15 5d cf 87 0d ......Hyper-V Virtual Ethernet Adapter
  8...9e ad 97 c7 60 2d ......Microsoft Wi-Fi Direct Virtual Adapter
 17...9e ad 97 c7 68 2d ......Microsoft Wi-Fi Direct Virtual Adapter #2
  6...6c c2 17 76 f7 51 ......Realtek PCIe FE Family Controller
  5...9c ad 97 c7 60 2d ......Broadcom BCM43142 802.11 bgn Wi-Fi Adapter
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.1.1      192.168.1.8     35
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      172.29.96.0    255.255.240.0         On-link       172.29.96.1    271
      172.29.96.1  255.255.255.255         On-link       172.29.96.1    271
  172.29.111.255  255.255.255.255         On-link       172.29.96.1    271
      192.168.1.0    255.255.255.0         On-link       192.168.1.8    291
      192.168.1.8  255.255.255.255         On-link       192.168.1.8    291
    192.168.1.255  255.255.255.255         On-link       192.168.1.8    291
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link       192.168.1.8    291
        224.0.0.0        240.0.0.0         On-link       172.29.96.1    271
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link       192.168.1.8    291
  255.255.255.255  255.255.255.255         On-link       172.29.96.1    271
===========================================================================
Persistent Routes:
  None

IPv6 Route Table
===========================================================================
Active Routes:
 If Metric Network Destination      Gateway
  1    331 ::1/128                  On-link
  6    291 fe80::/64                On-link
 50    271 fe80::/64                On-link
 50    271 fe80::131f:8a90:6485:4e87/128
```

2. Study of Network-related Commands(Linux)

Ifconfig:

```
aking@Amans-MacBook-Air ~ % ifconfig
lo0: flags=8049<UP,LOOPBACK,RUNNING,MULTICAST> mtu 16384
        options=1203<RXCSUM,TXCSUM,TXSTATUS,SW_TIMESTAMP>
        inet 127.0.0.1 netmask 0xff000000
        inet6 ::1 prefixlen 128
        inet6 fe80::1%lo0 prefixlen 64 scopeid 0x1
        nd6 options=201<PERFORMNUD,DAD>
gif0: flags=8010<POINTOPOINT,MULTICAST> mtu 1280
stf0: flags=0<> mtu 1280
anpi0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 9a:e1:b9:6c:e9:49
        nd6 options=201<PERFORMNUD,DAD>
        media: 100baseTX <full-duplex>
        status: inactive
anpi1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 9a:e1:b9:6c:e9:4a
        media: none
        status: inactive
en3: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 9a:e1:b9:6c:e9:29
        nd6 options=201<PERFORMNUD,DAD>
        media: 100baseTX <full-duplex>
        status: inactive
en4: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 9a:e1:b9:6c:e9:2a
        nd6 options=201<PERFORMNUD,DAD>
        media: none
        status: inactive
en1: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
        options=460<TSO4,TSO6,CHANNEL_IO>
        ether 36:c1:b4:72:9e:c0
        media: autoselect <full-duplex>
        status: inactive
en2: flags=8963<UP,BROADCAST,SMART,RUNNING,PROMISC,SIMPLEX,MULTICAST> mtu 1500
        options=460<TSO4,TSO6,CHANNEL_IO>
        ether 36:c1:b4:72:9e:c4
        media: autoselect <full-duplex>
        status: inactive
bridge0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=63<RXCSUM,TXCSUM,TSO4,TSO6>
        ether 36:c1:b4:72:9e:c0
        Configuration:
                id 0:0:0:0:0:0 priority 0 hellotime 0 fwddelay 0
                maxage 0 holdcnt 0 proto stp maxaddr 100 timeout 1200
                root id 0:0:0:0:0:0 priority 0 ifcost 0 port 0
                ipfilter disabled flags 0x0
        member: en1 flags=3<LEARNING,DISCOVER>
                ifmaxaddr 0 port 8 priority 0 path cost 0
        member: en2 flags=3<LEARNING,DISCOVER>
                ifmaxaddr 0 port 9 priority 0 path cost 0
        nd6 options=201<PERFORMNUD,DAD>
        media: <unknown type>
        status: inactive
ap1: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether 32:6a:6c:ac:49:22
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect (none)
        status: inactive
en0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether 86:ce:8b:f8:1c:c5
        inet6 fe80::4d8:33f9:b851:5a4d%en0 prefixlen 64 secured scopeid 0xb
        inet 192.168.1.7 netmask 0xffffff00 broadcast 192.168.1.255
        inet6 2401:4900:1c5b:402c:1017:133f:a8ed:dbc4 prefixlen 64 autoconf secured
        inet6 2401:4900:1c5b:402c:c95d:7e2:2899:34fd prefixlen 64 autoconf temporary
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
```

```
awdl0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=6460<TSO4,TSO6,CHANNEL_IO,PARTIAL_CSUM,ZEROINVERT_CSUM>
        ether 1a:1f:71:ba:7e:0e
        inet6 fe80::181f:71ff:feba:7e0e%awdl0 prefixlen 64 scopeid 0xd
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect
        status: active
llw0: flags=8863<UP,BROADCAST,SMART,RUNNING,SIMPLEX,MULTICAST> mtu 1500
        options=400<CHANNEL_IO>
        ether 1a:1f:71:ba:7e:0e
        inet6 fe80::181f:71ff:feba:7e0e%llw0 prefixlen 64 scopeid 0xe
        nd6 options=201<PERFORMNUD,DAD>
        media: autoselect (none)
utun0: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1500
        inet6 fe80::c436:9b97:aee8:cf2d%utun0 prefixlen 64 scopeid 0xf
        nd6 options=201<PERFORMNUD,DAD>
utun1: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1380
        inet6 fe80::7d70:6f91:27b7:127e%utun1 prefixlen 64 scopeid 0x10
        nd6 options=201<PERFORMNUD,DAD>
utun2: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 2000
        inet6 fe80::4205:72eb:2387:f346%utun2 prefixlen 64 scopeid 0x11
        nd6 options=201<PERFORMNUD,DAD>
utun3: flags=8051<UP,POINTOPOINT,RUNNING,MULTICAST> mtu 1000
        inet6 fe80::ce81:b1c:bd2c:69e%utun3 prefixlen 64 scopeid 0x12
        nd6 options=201<PERFORMNUD,DAD>
```

Ping google.com

```
aking@Amans-MacBook-Air ~ % ping google.com
PING google.com (172.217.24.78): 56 data bytes
64 bytes from 172.217.24.78: icmp_seq=0 ttl=118 time=9.201 ms
64 bytes from 172.217.24.78: icmp_seq=1 ttl=118 time=6.851 ms
64 bytes from 172.217.24.78: icmp_seq=2 ttl=118 time=8.798 ms
64 bytes from 172.217.24.78: icmp_seq=3 ttl=118 time=8.527 ms
64 bytes from 172.217.24.78: icmp_seq=4 ttl=118 time=6.760 ms
64 bytes from 172.217.24.78: icmp_seq=5 ttl=118 time=8.709 ms
64 bytes from 172.217.24.78: icmp_seq=6 ttl=118 time=7.669 ms
64 bytes from 172.217.24.78: icmp_seq=7 ttl=118 time=7.166 ms
^C
--- google.com ping statistics ---
8 packets transmitted, 8 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 6.760/7.960/9.201/0.902 ms
```

traceroute openai.com

```
aking@Amans-MacBook-Air ~ % traceroute openai.com
traceroute: Warning: openai.com has multiple addresses; using 172.64.154.211
traceroute to openai.com (172.64.154.211), 64 hops max, 40 byte packets
 1  192.168.1.1 (192.168.1.1)  3.127 ms  2.514 ms  2.882 ms
 2  abts-north-static-068.127.176.122.airtelbroadband.in (122.176.127.68)  3.962 ms  5.758 ms  3.973 ms
 3  182.79.117.229 (182.79.117.229)  3.071 ms  6.195 ms
    182.79.117.225 (182.79.117.225)  3.913 ms
 4  182.79.235.253 (182.79.235.253)  5.629 ms
    116.119.73.221 (116.119.73.221)  5.338 ms
    182.79.235.253 (182.79.235.253)  6.310 ms
```

netstat -tulnp

```
aking@Amans-MacBook-Air ~ % netstat tuln
Active Internet connections
Proto Recv-Q Send-Q  Local Address          Foreign Address        (state)
tcp6       0      0  2401:4900:1c5b:4.59797 sa-in-xbc.1e100..5228  ESTABLISHED
tcp6       0      0  2401:4900:1c5b:4.56947 whatsapp-chatd-e.443   ESTABLISHED
tcp4       0      0  192.168.1.7.59753      17.242.13.4.443         ESTABLISHED
udp6       0      0  2401:4900:1c5b:4.57615 del11s15-in-x0e..https
udp6       0      0  2401:4900:1c5b:4.56767 del11s05-in-x0e..https
udp6       0      0  2401:4900:1c5b:4.59310 tzdela-aq-in-x0e.https
udp6       0      0  2401:4900:1c5b:4.61236 *.*
udp4       0      0  192.168.1.7.55805      *.*
udp46      0      0  *.53942                *.*
udp4       0      0  *.59600                *.*
udp46      0      0  *.mdns                 *.*
udp46      0      0  *.mdns                 *.*
udp46      0      0  *.mdns                 *.*
udp46      0      0  *.mdns                 *.*
udp46      0      0  *.mdns                 *.*
udp46      0      0  *.mdns                 *.*
udp46      0      0  *.mdns                 *.*
udp4       0      0  *.mdns                 *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp46      0      0  *.65262                *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp46      0      0  *.*                    *.*
udp4       0      0  *.*                    *.*
udp6       0      0  *.mdns                 *.*
udp4       0      0  *.mdns                 *.*
udp4       0      0  *.netbios-ns           *.*
udp4       0      0  *.netbios-dgm          *.*
Active Multipath Internet connections
Proto/ID  Flags      Local Address          Foreign Address        (state)
icm6       0      0  *.*                    *.*
Active LOCAL (UNIX) domain sockets
Address          Type   Recv-Q Send-Q            Inode             Conn              Refs          Nextref Addr
42a157c9f860d110 stream      0      0                0 8f1514b9c5eb8711                0               0 /var/run/mDNSResponder
8f1514b9c5eb8711 stream      0      0                0 42a157c9f860d110                0               0
8cd72618dcd64132 stream      0      0                0 b79914b646bd0adc                0               0
b79914b646bd0adc stream      0      0                0 8cd72618dcd64132                0               0
73fa73a837267c9b stream      0      0                0 61c2c15ba1a80550                0               0
61c2c15ba1a80550 stream      4      0                0 73fa73a837267c9b                0               0
3efbf60e82fdea71 stream      0      0                0    353c47f3fb26e                0               0
   353c47f3fb26e stream      0      0                0 3efbf60e82fdea71                0               0
485c3e0c0829ca91 stream      0      0                0 451024d95269c8d4                0               0 /var/run/mDNSResponder
451024d95269c8d4 stream      0      0                0 485c3e0c0829ce91                0               0
1633fed769206de7 stream      0      0                0 41cc07badabdf510                0               0 /var/run/mDNSResponder
41cc07badabdf510 stream      0      0                0 1633fed769206de7                0               0
be5dbdf85a1c6653 stream      0      0                0 737fb927f2ca095d                0               0 /var/run/mDNSResponder
737fb927f2ca095d stream      0      0                0 be5dbdf85a1c6653                0               0
5c8ea13451e6f4a9 stream      0      0                0 ed4fe27e0a203dd8                0               0 /var/run/mDNSResponder
ed4fe27e0a203dd8 stream      0      0                0 5c8ea13451e6f4a9                0               0
abf95eb8320e4982 stream      0      0                0 c3fb928e3a662271                0               0 /var/run/mDNSResponder
c3fb928e3a662271 stream      0      0                0 abf95eb8320e4982                0               0
b5175cd012f08235 stream      0      0                0 97d7fef76c5c6125                0               0 /var/run/mDNSResponder
97d7fef76c5c6125 stream      0      0                0 b5175cd012f08235                0               0
```

arp -a

```
aking@Amans-MacBook-Air ~ % arp -a
? (192.168.1.1) at fc:9f:2a:2a:7d:79 on en0 ifscope [ethernet]
mdns.mcast.net (224.0.0.251) at 1:0:5e:0:0:fb on en0 ifscope permanent [ethernet]
```

nslookup google.com

```
aking@Amans-MacBook-Air ~ % nslookup google.com
Server:         2401:4900:50:9::7dd
Address:        2401:4900:50:9::7dd#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.24.78
```

Dig google.com

```
aking@Amans-MacBook-Air ~ % dig ggogle.com

; <<>> DiG 9.10.6 <<>> ggogle.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 12644
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;ggogle.com.                    IN      A

;; AUTHORITY SECTION:
ggogle.com.             300     IN      SOA     ns1.googledomains.com. cloud-dns-hostmaster.google.com. 1 21600 3600 259200 300

;; Query time: 181 msec
;; SERVER: 2401:4900:50:9::7dd#53(2401:4900:50:9::7dd)
;; WHEN: Sat Nov 22 19:28:12 IST 2025
;; MSG SIZE  rcvd: 121
```

3. Analysis of Windows registry.

Steps:

1.Search for registry Editor in search tab.

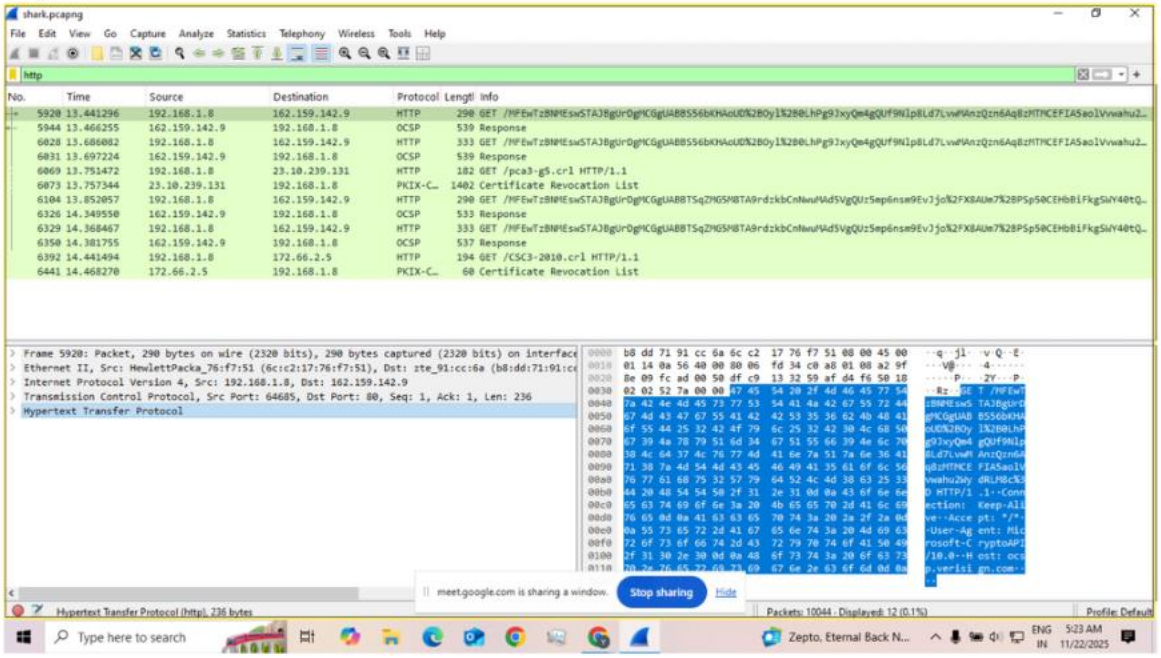
2.Inside The Registory Editor, Go to Computer then search for **HKEY_CURRENT_USER (HKCU)** .

3.**HKCU\Software** → user software settings

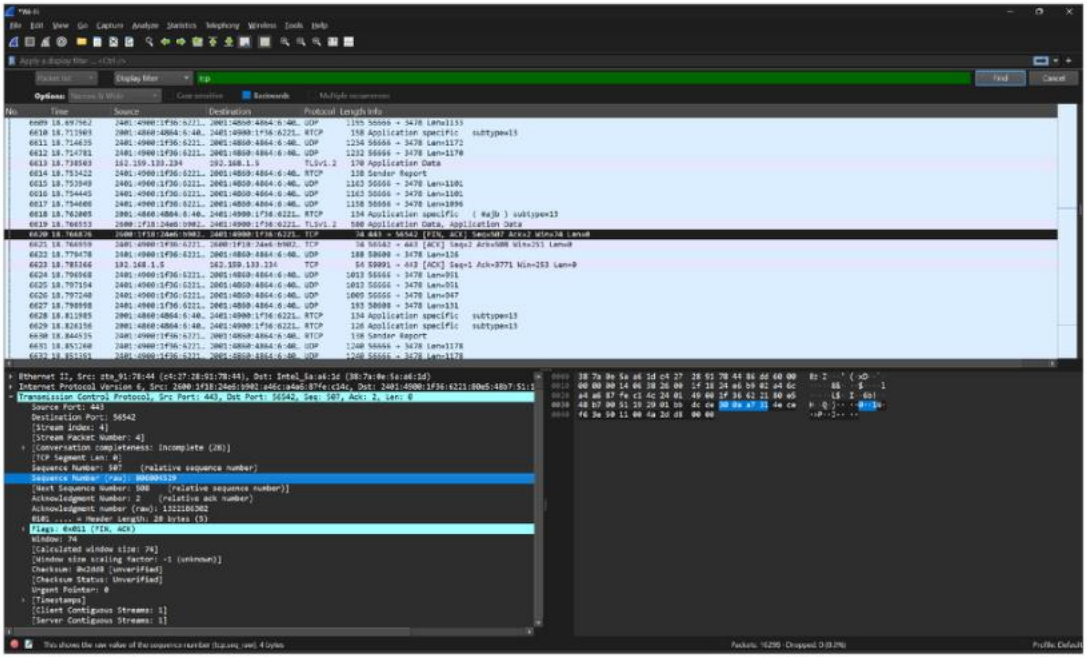
3.**HKCU\Software\Microsoft\Windows\CurrentVersion\Run** → startup programs

4. Capture and analyse network packets using Wireshark. Analyse the packets captured.

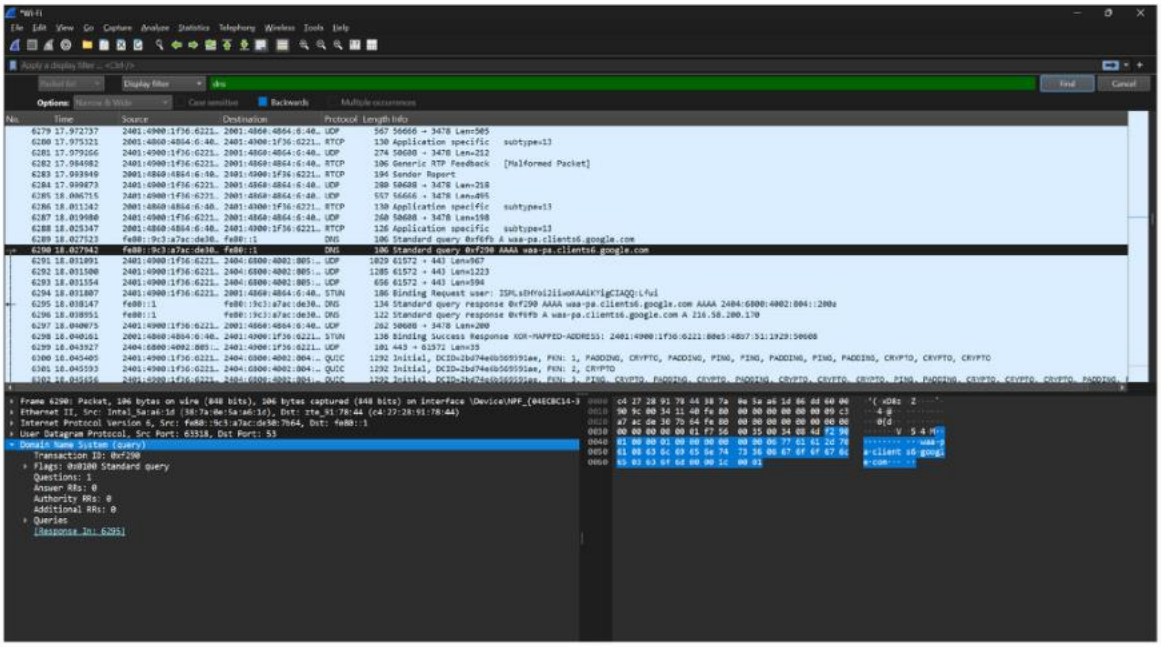
Http:



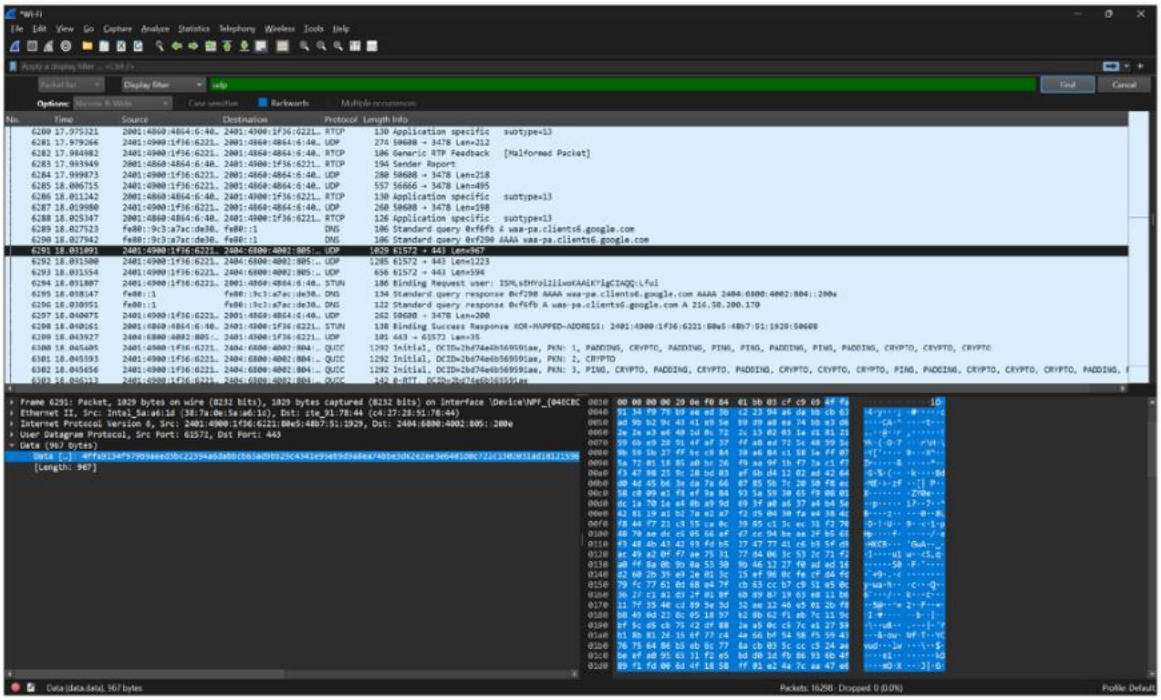
TCP



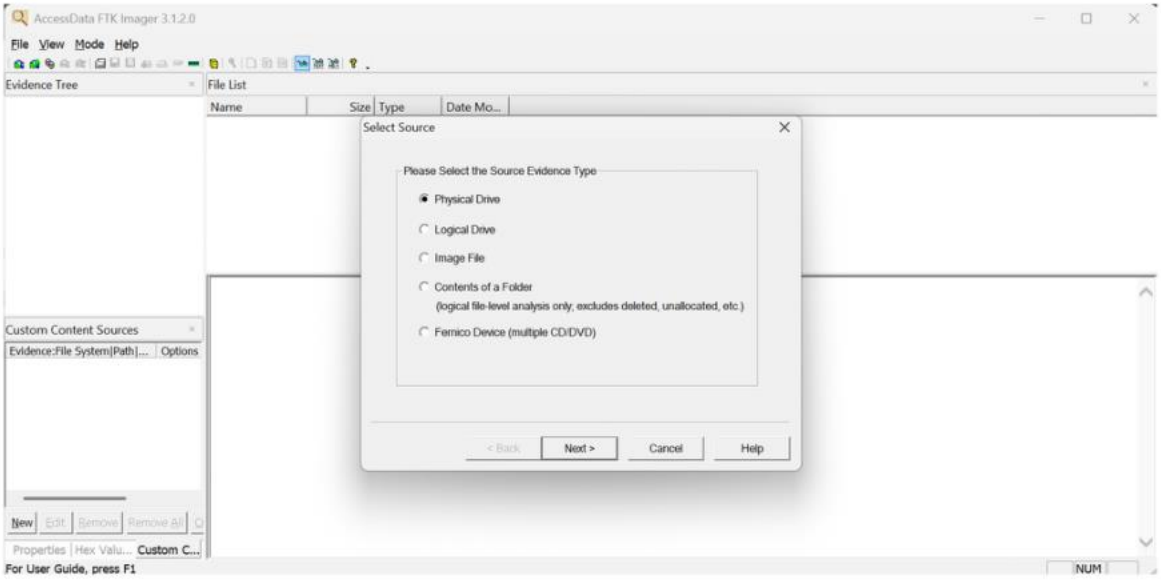
DNS:

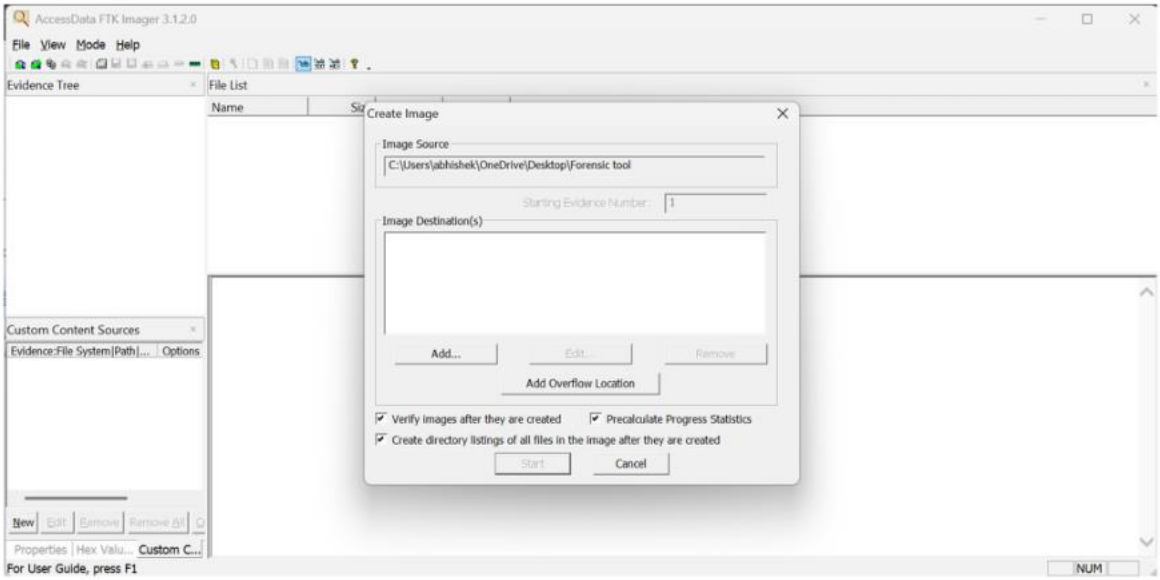


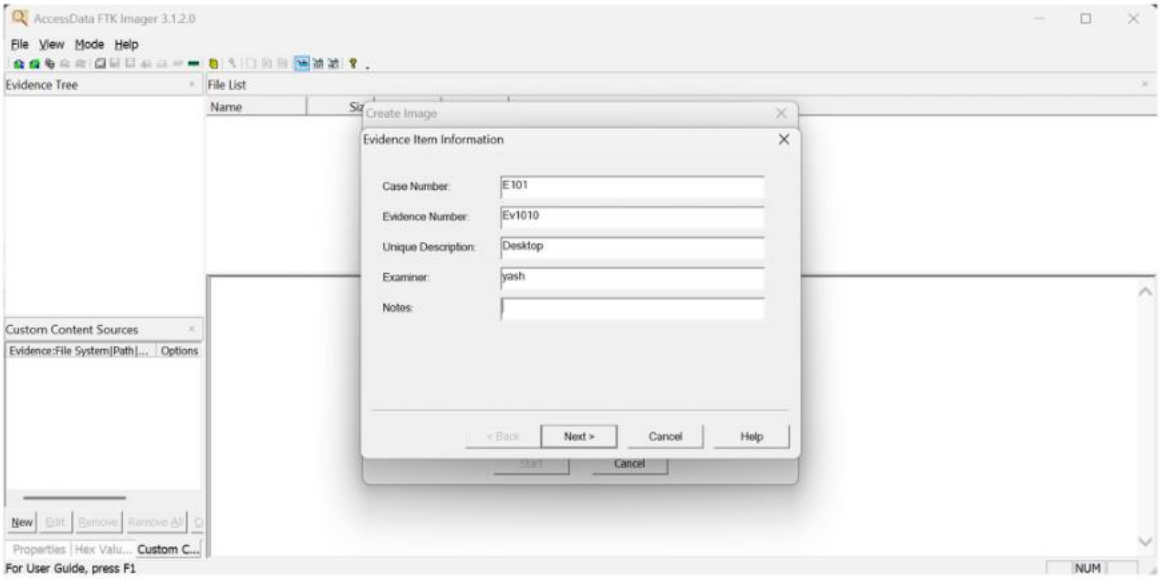
UDP:

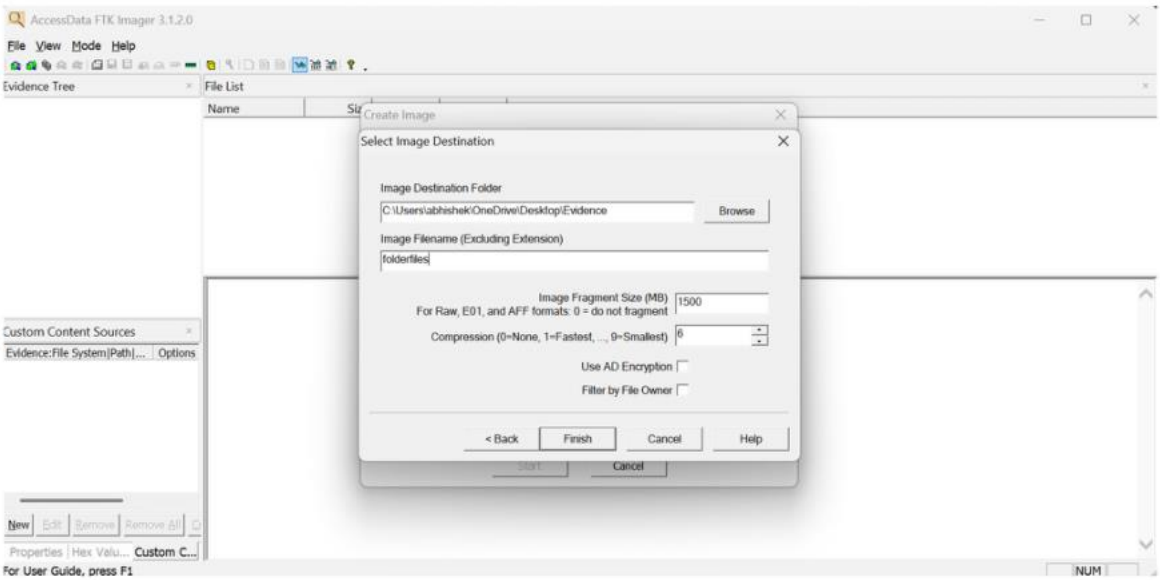


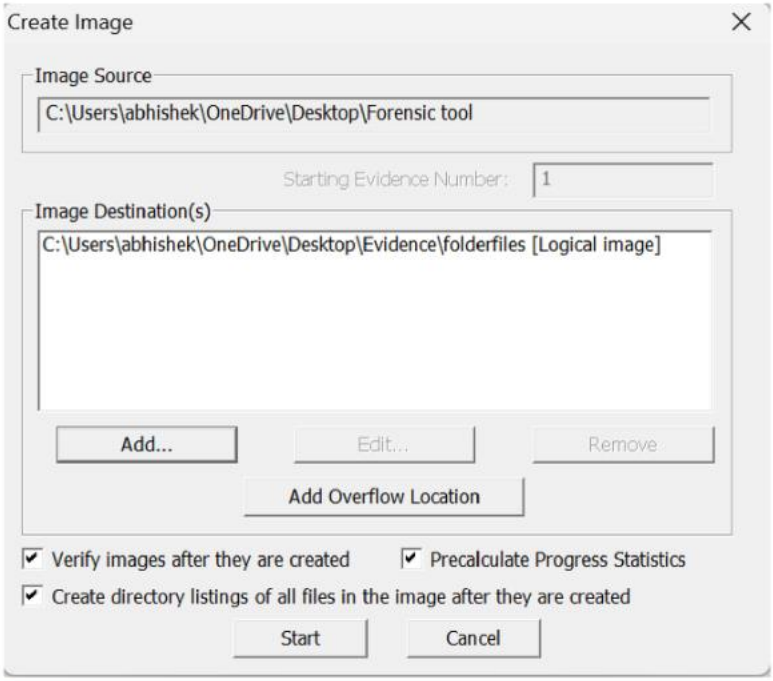
5. Creating a Forensic image using FTK Imager/ Encase Imager: creating a forensic image, checking the integrity of data, and analysing the forensic image.

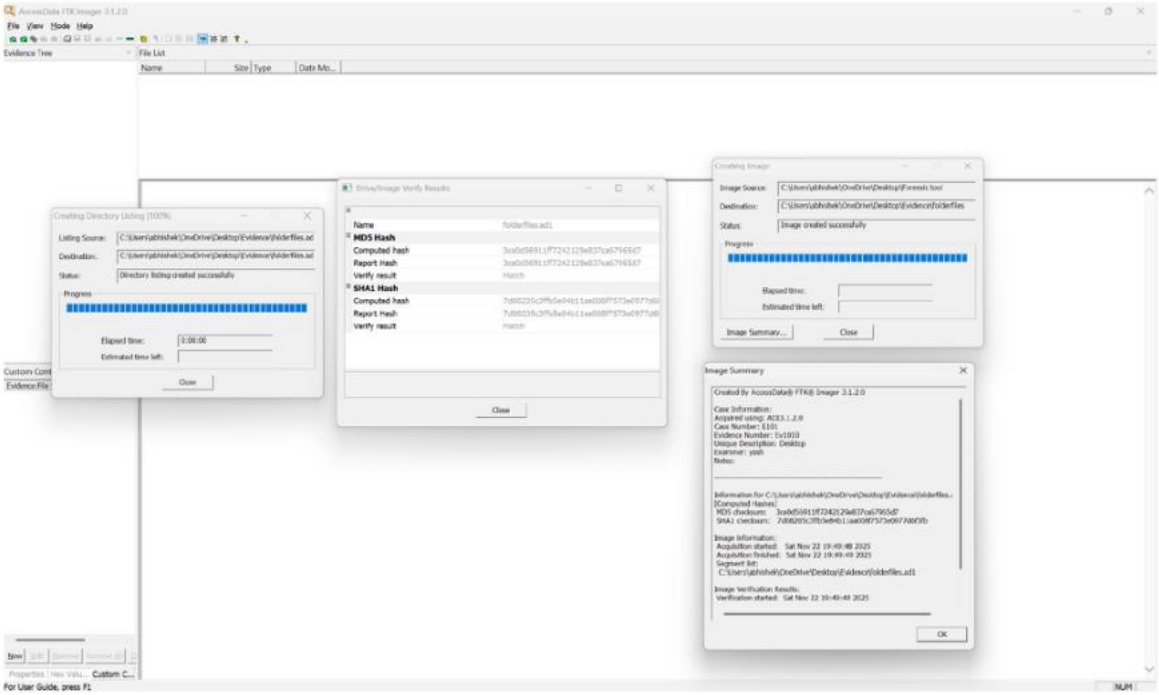




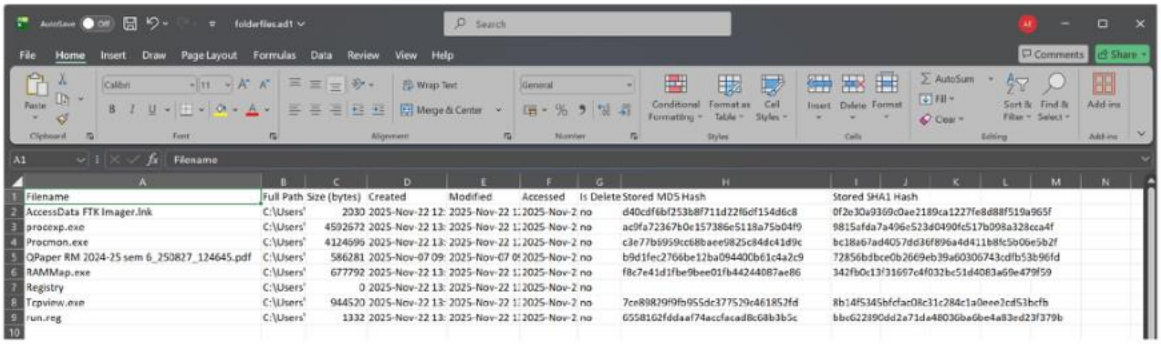

AccessData FTK Imager 3.1.2.0
File View Mode Help
Evidence Tree
File List
Name
Create Image
Evidence Item Information
Case Number: E101
Evidence Number: Ev1010
Unique Description: Desktop
Examiner: yash
Notes:
< Back
Next >
Cancel
Help
Start
Cancel
Custom Content Sources
Evidence:File System|Path|...
Options
New
Edit
Remove
Remove All
Properties
Hex Valu...
Custom C...
For User Guide, press F1
NUM


AccessData FTK Imager 3.1.2.0
File View Mode Help
Evidence Tree
File List
Name
Create Image
Select Image Destination
Image Destination Folder
C:\Users\abhishek\OneDrive\Desktop\Evidence
Browse
Image Filename (Excluding Extension)
folderfiles
Image Fragment Size (MB)
For Raw, E01, and AFF formats: 0 = do not fragment
1500
Compression (0=None, 1=Fastest, ..., 9=Smallest)
6
Use AD Encryption
Filter by File Owner
< Back
Finish
Cancel
Help
Start
Cancel
Custom Content Sources
Evidence:File System|Path|...
Options
New
Edit
Remove
Remove All
Properties
Hex Valu...
Custom C...
For User Guide, press F1
NUM

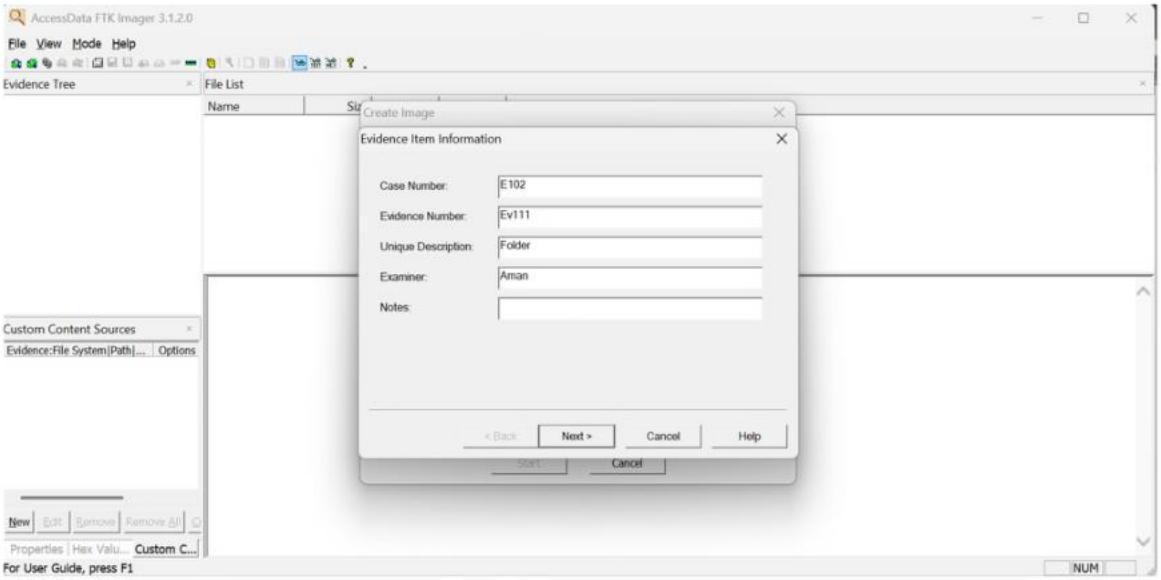

Create Image
Image Source
C:\Users\abhishek\OneDrive\Desktop\Forensic tool
Starting Evidence Number:
1
Image Destination(s)
C:\Users\abhishek\OneDrive\Desktop\Evidence\folderfiles [Logical image]
Add...
Edit...
Remove
Add Overflow Location
Verify images after they are created
Precalculate Progress Statistics
Create directory listings of all files in the image after they are created
Start
Cancel



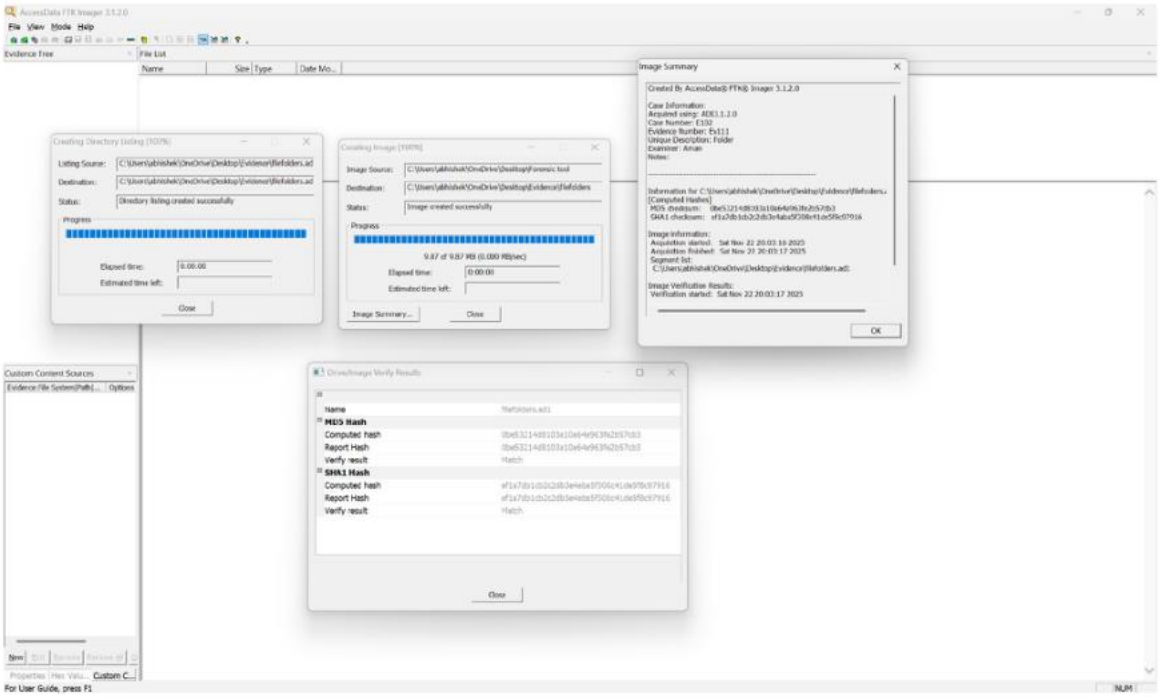
```
Created By AccessData® FTK® Imager 3.1.2.0

Case Information:
Acquired using: ADI3.1.2.0
Case Number: E101
Evidence Number: Ev1010
Unique Description: Desktop
Examiner: yash
Notes:

--------------------------------------------------------------

Information for C:\Users\abhishek\OneDrive\Desktop\Evidence\folderfiles.ad1:
[Computed Hashes]
 MD5 checksum:    3ce0d56911ff7242129e837ca67965d7
 SHA1 checksum:   7d08205c3ffb5e84b11aa008f7573e0977d6f5fb

Image information:
 Acquisition started:   Sat Nov 22 19:49:48 2025
 Acquisition finished:  Sat Nov 22 19:49:49 2025
 Segment list:
  C:\Users\abhishek\OneDrive\Desktop\Evidence\folderfiles.ad1

Image Verification Results:
 Verification started:  Sat Nov 22 19:49:49 2025
 Verification finished: Sat Nov 22 19:49:49 2025
 MD5 checksum:    3ce0d56911ff7242129e837ca67965d7 : verified
 SHA1 checksum:   7d08205c3ffb5e84b11aa008f7573e0977d6f5fb : verified
```



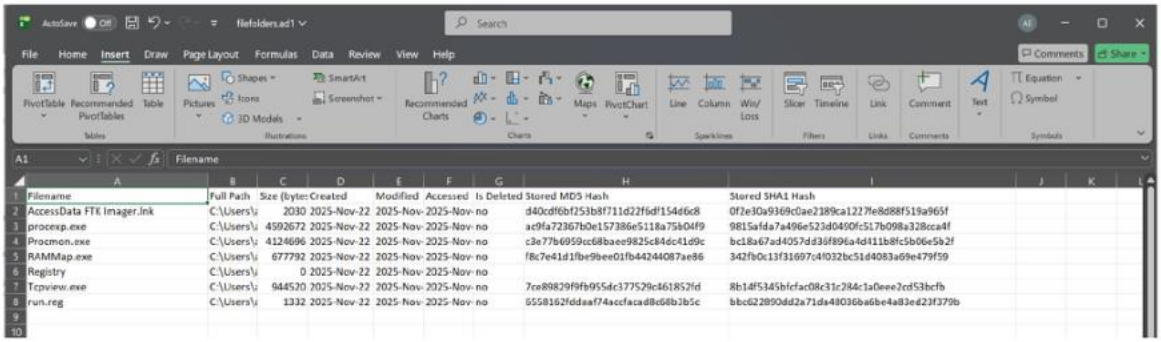
Filename	Full Path	Size (bytes)	Created	Modified	Accessed	Is Delete	Stored MD5 Hash	Stored SHA1 Hash
AccessData FTK Imager.lnk	C:\Users'	2030	2025-Nov-22 12:	2025-Nov-22 1	2025-Nov-2	no	d40cdf6bf253b8f711d22f6df154d6c8	0f2e30a9369c0ae2189ca1227fe8d88f519a965f
procexp.exe	C:\Users'	4592672	2025-Nov-22 13:	2025-Nov-22 1	2025-Nov-2	no	ac9fa72367b0e157386e5118a75b04f9	9815afda7a496e523d0490fc517b098a328cca4f
Procmon.exe	C:\Users'	4124696	2025-Nov-22 13:	2025-Nov-22 1	2025-Nov-2	no	c3e77b6959cc68bbaee9825c84dc41d9c	bc18a67ad4057dd36f896a4d411b8fc5b06e5b2f
QPaper RM 2024-25 sem 6_250827_124645.pdf	C:\Users'	586281	2025-Nov-07 09:	2025-Nov-07 0	2025-Nov-2	no	b9d1fec2766be12ba09440b61c4a2c9	72856bdbce0b2669eb39a60306743cdfb53b96fd
RAMMap.exe	C:\Users'	677792	2025-Nov-22 13:	2025-Nov-22 1	2025-Nov-2	no	f8c7e41d1fbe9bee01fb44244087ae86	342fb0c13f31697c4f032bc51d4083a69e479f59
Registry	C:\Users'	0	2025-Nov-22 13:	2025-Nov-22 1	2025-Nov-2	no		
Tcpview.exe	C:\Users'	944520	2025-Nov-22 13:	2025-Nov-22 1	2025-Nov-2	no	7ce898299fb955dc377529c46185fd	8b14f5345bfcfac08c31c284c1a0eee2cd53bcfb
run.reg	C:\Users'	1332	2025-Nov-22 13:	2025-Nov-22 1	2025-Nov-2	no	655816fddaaf74accfacad8c68b3b5c	bbc622890dd2a71da48036ba6be4a83ed23f379b



After deletion of file:



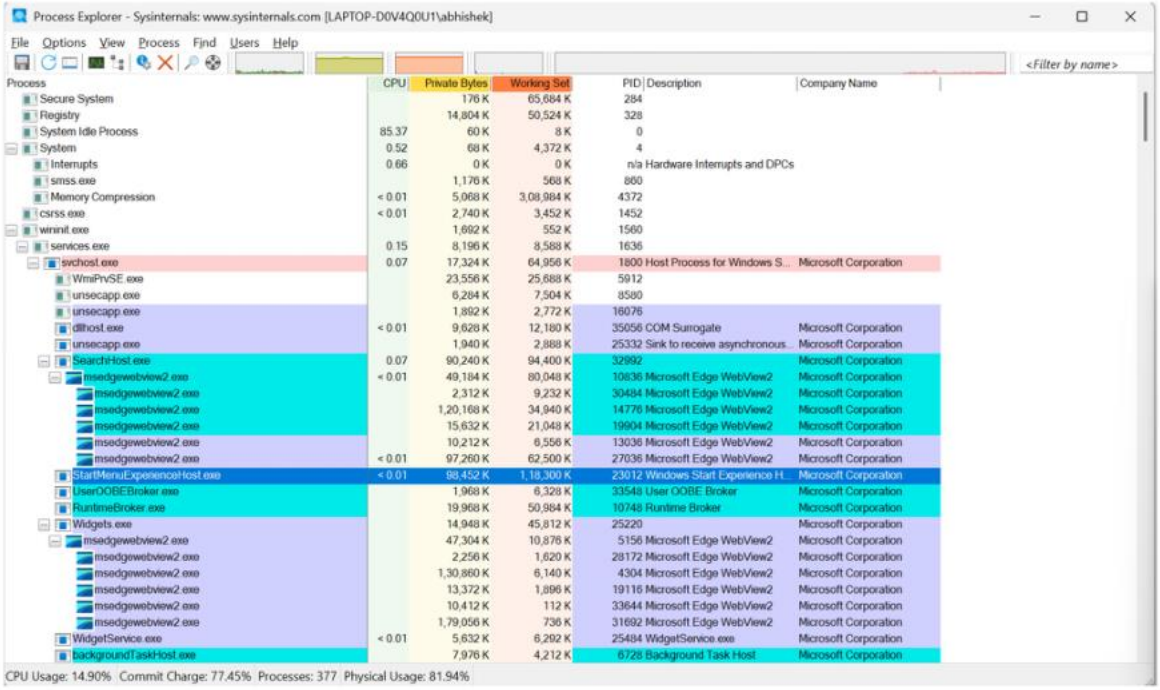
```
Created By AccessData® FTK® Imager 3.1.2.0

Case Information:
Acquired using: ADI3.1.2.0
Case Number: E102
Evidence Number: Ev111
Unique Description: Folder
Examiner: Aman
Notes:

---------------------------------------------------------------

Information for C:\Users\abhishek\OneDrive\Desktop\Evidence\filefolders.ad1:
[Computed Hashes]
 MD5 checksum:    0be53214d8103a10a64a963fe2b57cb3
 SHA1 checksum:   ef1a7db1cb2c2db3e4aba5f308c41de5f8c97916

Image information:
 Acquisition started:   Sat Nov 22 20:03:16 2025
 Acquisition finished:  Sat Nov 22 20:03:17 2025
 Segment list:
  C:\Users\abhishek\OneDrive\Desktop\Evidence\filefolders.ad1

Image Verification Results:
 Verification started:  Sat Nov 22 20:03:17 2025
 Verification finished: Sat Nov 22 20:03:17 2025
 MD5 checksum:    0be53214d8103a10a64a963fe2b57cb3 : verified
 SHA1 checksum:   ef1a7db1cb2c2db3e4aba5f308c41de5f8c97916 : verified
```



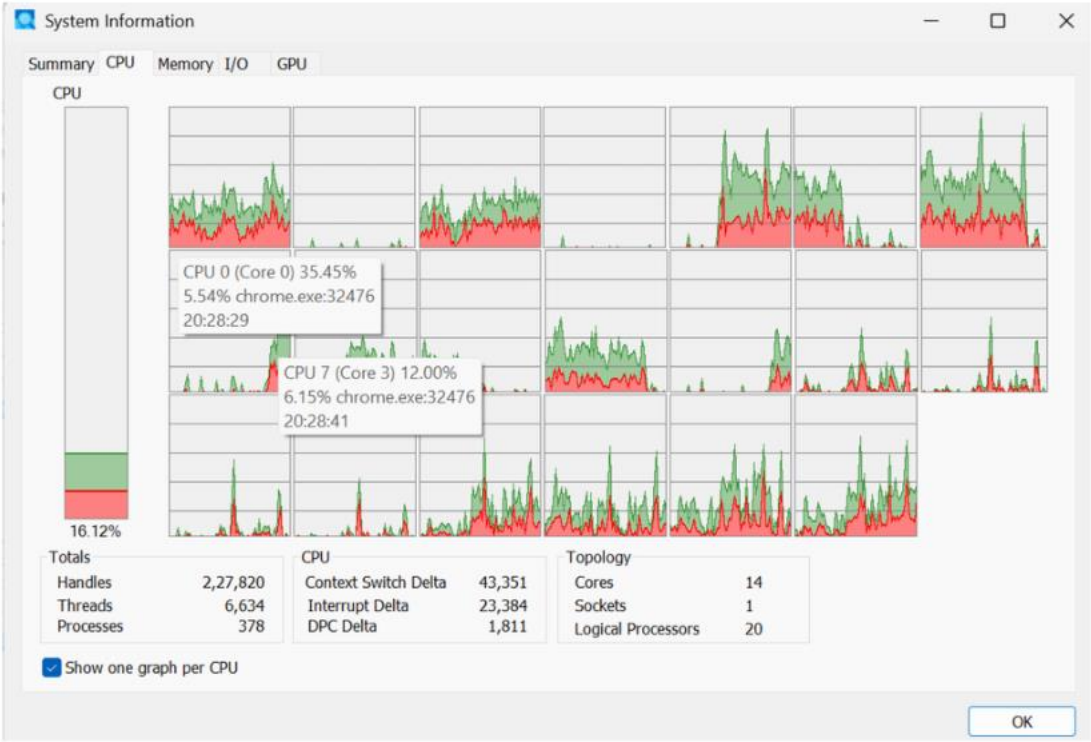
Filename	Full Path	Size (byte:	Created	Modified	Accessed	Is Deleted	Stored MD5 Hash	Stored SHA1 Hash
AccessData FTK Imager.lnk	C:\Users\	2030	2025-Nov-22	2025-Nov-	2025-Nov-	no	d40cdf6bf253b8f711d22f6df154d6c8	0f2e30a9369c0ae2189ca1227fe8d88f519a965f
procexp.exe	C:\Users\	4592672	2025-Nov-22	2025-Nov-	2025-Nov-	no	ac9fa72367b0e157386e5118a75b04f9	9815afda7a496e523d0490fc517b098a328cca4f
Procmon.exe	C:\Users\	4124696	2025-Nov-22	2025-Nov-	2025-Nov-	no	c3e77b6959cc68baee9825c84dc41d9c	bc18a67ad4057dd36f896a4d411b8fc5b06e5b2f
RAMMap.exe	C:\Users\	677792	2025-Nov-22	2025-Nov-	2025-Nov-	no	f8c7e41d1fbe9bee01fb44244087ae86	342fb0c13f31697c4f032bc51d4083a69e479f59
Registry	C:\Users\	0	2025-Nov-22	2025-Nov-	2025-Nov-	no		
Tcpview.exe	C:\Users\	944520	2025-Nov-22	2025-Nov-	2025-Nov-	no	7ce89829f9fb955dc377529c461852fd	8b14f5345bfcfac08c31c284c1a0eee2cd53bcfb
run.reg	C:\Users\	1332	2025-Nov-22	2025-Nov-	2025-Nov-	no	6558162fddaaf74accfacad8c68b2b5c	bbc622890dd2a71da48036ba6be4a83ed23f379b

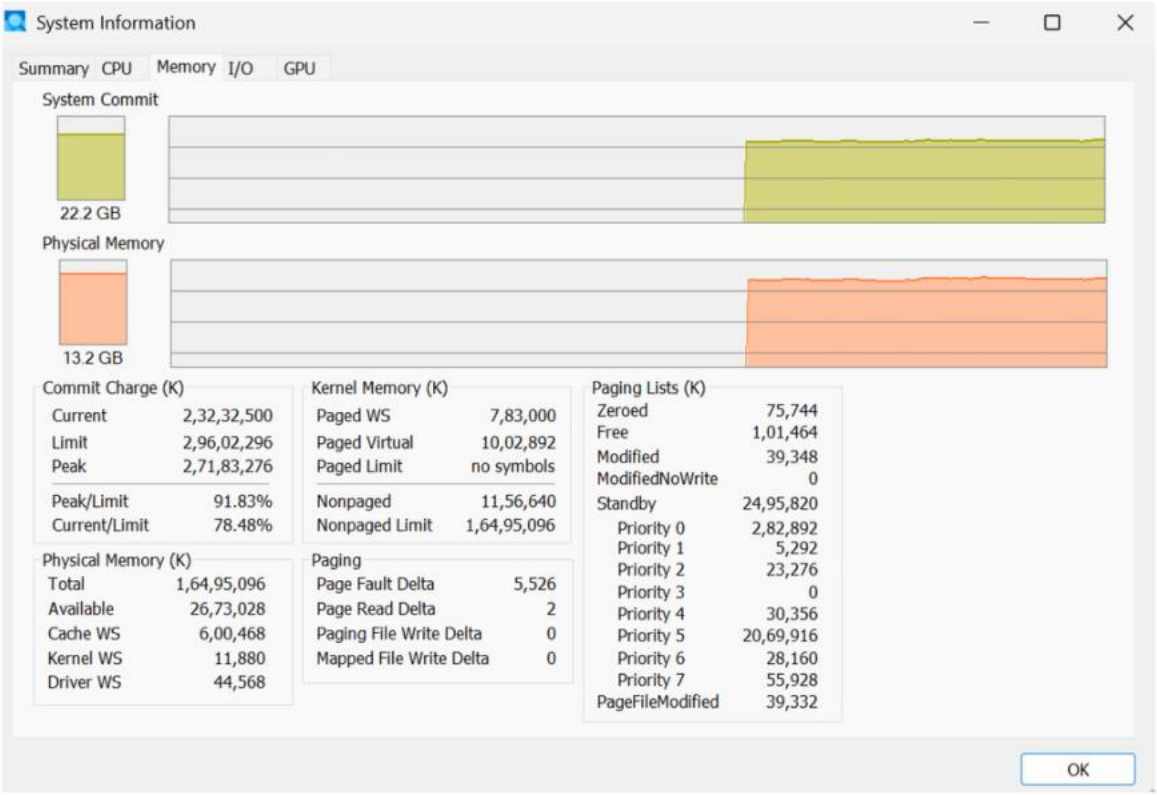
6. Using System internal tools for network tracking and process monitoring, do the following:

a. Monitor live processes

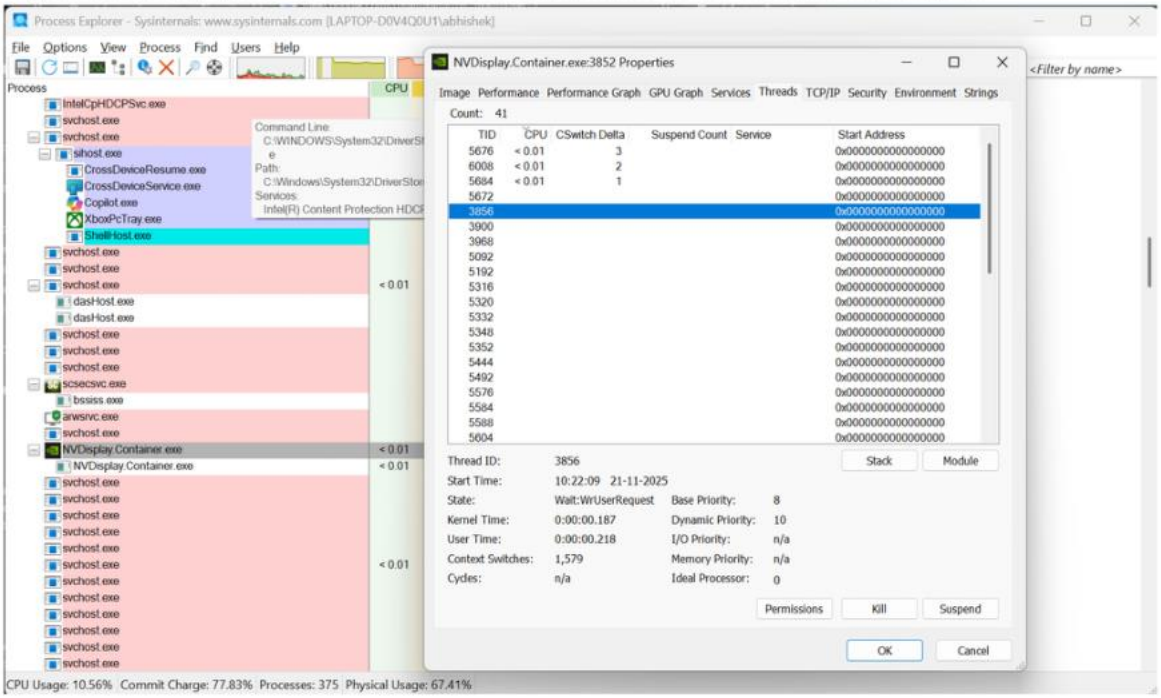


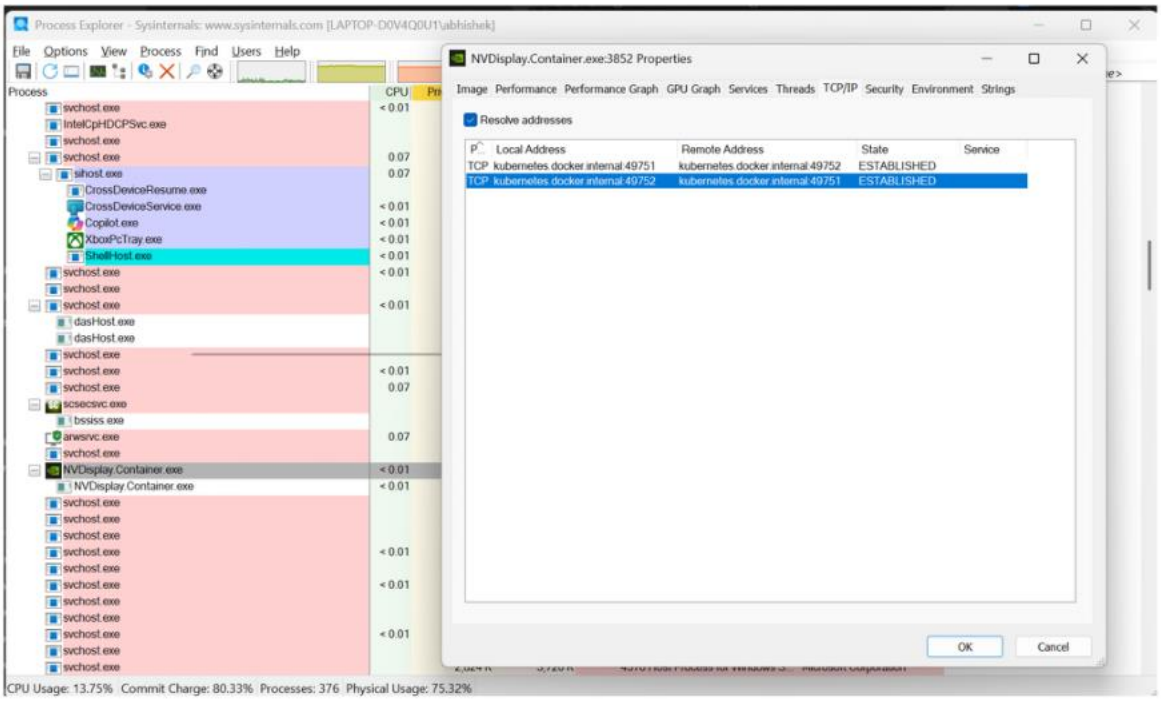
Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System		176 K	65,684 K	284		
Registry		14,804 K	50,524 K	328		
System Idle Process	85.37	60 K	8 K	0		
System	0.52	68 K	4,372 K	4		
Interrupts	0.66	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		1,176 K	568 K	860		
Memory Compression	< 0.01	5,068 K	3,08,984 K	4372		
csrss.exe	< 0.01	2,740 K	3,452 K	1452		
wininit.exe		1,692 K	552 K	1560		
services.exe	0.15	8,196 K	8,588 K	1636		
svchost.exe	0.07	17,324 K	64,956 K	1800	Host Process for Windows S...	Microsoft Corporation
WmiPrvSE.exe		23,556 K	25,688 K	5912		
unsecapp.exe		6,284 K	7,504 K	8580		
unsecapp.exe		1,892 K	2,772 K	16076		
dllhost.exe	< 0.01	9,628 K	12,180 K	35056	COM Surrogate	Microsoft Corporation
unsecapp.exe		1,940 K	2,888 K	25332	Sink to receive asynchronous...	Microsoft Corporation
SearchHost.exe	0.07	90,240 K	94,400 K	32992		Microsoft Corporation
msedgewebview2.exe	< 0.01	49,184 K	80,048 K	10836	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		2,312 K	9,232 K	30484	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		1,20,168 K	34,940 K	14776	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		15,632 K	21,048 K	19904	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		10,212 K	6,556 K	13036	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe	< 0.01	97,260 K	62,500 K	27036	Microsoft Edge WebView2	Microsoft Corporation
StartMenuExperienceHost.exe	< 0.01	98,452 K	1,18,300 K	23012	Windows Start Experience H...	Microsoft Corporation
UserOOBEBroker.exe		1,968 K	6,328 K	33548	User OOBE Broker	Microsoft Corporation
RuntimeBroker.exe		19,968 K	50,984 K	10748	Runtime Broker	Microsoft Corporation
Widgets.exe		14,948 K	45,812 K	25220		Microsoft Corporation
msedgewebview2.exe		47,304 K	10,876 K	5156	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		2,256 K	1,620 K	28172	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		1,30,860 K	6,140 K	4304	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		13,372 K	1,896 K	19116	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		10,412 K	112 K	33644	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		1,79,056 K	736 K	31692	Microsoft Edge WebView2	Microsoft Corporation
WidgetService.exe	< 0.01	5,632 K	6,292 K	25484	WidgetService.exe	Microsoft Corporation
backgroundTaskHost.exe		7,976 K	4,212 K	6728	Background Task Host	Microsoft Corporation


System Information
Summary CPU Memory I/O GPU
CPU
16.12%
CPU 0 (Core 0) 35.45%
5.54% chrome.exe:32476
20:28:29
CPU 7 (Core 3) 12.00%
6.15% chrome.exe:32476
20:28:41
Totals
Handles 2,27,820
Threads 6,634
Processes 378
CPU
Context Switch Delta 43,351
Interrupt Delta 23,384
DPC Delta 1,811
Topology
Cores 14
Sockets 1
Logical Processors 20
Show one graph per CPU
OK

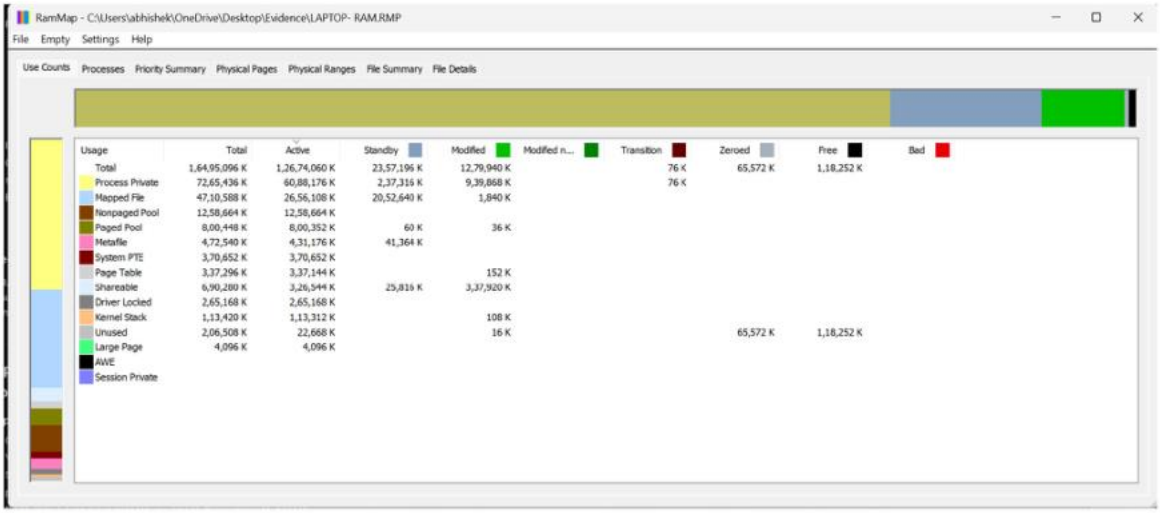

System Information
Summary CPU Memory I/O GPU
System Commit
22.2 GB
Physical Memory
13.2 GB
Commit Charge (K)
Current 2,32,32,500
Limit 2,96,02,296
Peak 2,71,83,276
Peak/Limit 91.83%
Current/Limit 78.48%
Kernel Memory (K)
Paged WS 7,83,000
Paged Virtual 10,02,892
Paged Limit no symbols
Nonpaged 11,56,640
Nonpaged Limit 1,64,95,096
Paging Lists (K)
Zeroed 75,744
Free 1,01,464
Modified 39,348
ModifiedNoWrite 0
Standby 24,95,820
Priority 0 2,82,892
Priority 1 5,292
Priority 2 23,276
Priority 3 0
Priority 4 30,356
Priority 5 20,69,916
Priority 6 28,160
Priority 7 55,928
PageFileModified 39,332
Physical Memory (K)
Total 1,64,95,096
Available 26,73,028
Cache WS 6,00,468
Kernel WS 11,880
Driver WS 44,568
Paging
Page Fault Delta 5,526
Page Read Delta 2
Paging File Write Delta 0
Mapped File Write Delta 0
OK

Right click on any process and in that check properties under that you can inspect.





b. Capture RAM



Usage	Total	Active	Standby	Modified	Modified n...	Transition	Zeroed	Free	Bad
Total	1,64,95,096 K	1,26,74,060 K	23,57,196 K	12,79,940 K		76 K	65,572 K	1,18,252 K	
Process Private	72,65,436 K	60,88,176 K	2,37,316 K	9,39,868 K		76 K			
Mapped File	47,10,588 K	26,56,108 K	20,52,640 K	1,840 K					
Nonpaged Pool	12,58,664 K	12,58,664 K							
Paged Pool	8,00,448 K	8,00,352 K	60 K	36 K					
Metafile	4,72,540 K	4,31,176 K	41,364 K						
System PTE	3,70,652 K	3,70,652 K							
Page Table	3,37,296 K	3,37,144 K		152 K					
Shareable	6,90,280 K	3,26,544 K	25,816 K	3,37,920 K					
Driver Locked	2,65,168 K	2,65,168 K							
Kernel Stack	1,13,420 K	1,13,312 K		108 K					
Unused	2,06,508 K	22,668 K		16 K			65,572 K	1,18,252 K	
Large Page	4,096 K	4,096 K							
AWE									
Session Private									

RamMap - C:\Users\abhishek\OneDrive\Desktop\Evidence\LAPTOP- RAM.RMP

File Empty Settings Help

Use Counts Processes Priority Summary Physical Pages Physical Ranges File Summary File Details

Process	Session	PID	Private	Standby	Modified	Page Table	Total
System	-1	4	0 K	1,408 K	0 K	88 K	1,496 K
Secure System	-1	284	0 K	8 K	0 K	48 K	56 K
Registry	-1	328	3,276 K	3,500 K	0 K	384 K	7,160 K
msedge.exe	4	512	0 K	2,624 K	0 K	36 K	2,660 K
WUDFHost.exe	0	840	128 K	1,868 K	0 K	252 K	2,248 K
smss.exe	-1	860	92 K	8 K	0 K	160 K	260 K
WindowsTer...	4	872	0 K	1,824 K	0 K	32 K	1,856 K
WUDFHost.exe	0	1032	140 K	1,628 K	0 K	332 K	2,100 K
SrTasks.exe	0	1124	0 K	8 K	0 K	32 K	40 K
msedgewebvi...	4	1256	11,288 K	1,628 K	0 K	1,288 K	14,204 K
csrss.exe	0	1452	1,004 K	0 K	0 K	296 K	1,300 K
Adobe Installe	1	1464	0 K	1,168 K	0 K	28 K	1,196 K
wininit.exe	0	1560	8 K	1,144 K	8 K	292 K	1,452 K
csrss.exe	1	1568	0 K	1,152 K	0 K	32 K	1,184 K
Tobii.EyeX.Eng	3	1584	0 K	1,136 K	0 K	32 K	1,168 K
services.exe	0	1636	3,976 K	8 K	156 K	436 K	4,576 K
RuntimeBroker.	4	1644	348 K	4 K	0 K	288 K	640 K
LsaIso.exe	0	1648	240 K	1,788 K	0 K	184 K	2,212 K
lsass.exe	0	1664	10,220 K	2,972 K	0 K	556 K	13,748 K
cmd.exe	4	1716	464 K	4 K	0 K	172 K	640 K
HPSUPD-Win...	2	1760	0 K	1,036 K	0 K	32 K	1,068 K
taskhostw.exe	0	1792	0 K	1,004 K	0 K	32 K	1,036 K
svchost.exe	0	1800	12,656 K	1,008 K	0 K	672 K	14,336 K
fontdrvhost.ex	0	1816	60 K	984 K	0 K	196 K	1,240 K
WUDFHost.exe	0	1896	636 K	900 K	0 K	424 K	1,960 K

c. Capture TCP/UDP packets

TCP

Process Monitor - C:\Users\yash\Desktop\Logfile.PML

File Edit Event Filter Tools Options Help

Time ...	Process Name	PID	Operation	Path	Result	Detail
5:05:0...	esrv.exe	9172	TCP Send	DESKTOP-SNEVEAD:64720 -> DESKT...	SUCCESS	Length: 41088, sta...
5:05:1...	svchost.exe	3544	TCP Send	DESKTOP-SNEVEAD:60541 -> 4.213.2...	SUCCESS	Length: 101, starti...
5:05:1...	svchost.exe	9296	TCP Send	DESKTOP-SNEVEAD:64721 -> 24.224....	SUCCESS	Length: 190, starti...
5:05:1...	svchost.exe	9296	TCP Send	DESKTOP-SNEVEAD:64721 -> 24.224....	SUCCESS	Length: 93, startim...
5:05:1...	svchost.exe	9296	TCP Send	DESKTOP-SNEVEAD:64721 -> 24.224....	SUCCESS	Length: 193, starti...
5:05:1...	MsMpEng.exe	3768	TCP Send	DESKTOP-SNEVEAD:64722 -> 57.155....	SUCCESS	Length: 203, starti...
5:05:1...	MsMpEng.exe	3768	TCP Send	DESKTOP-SNEVEAD:64722 -> 57.155....	SUCCESS	Length: 158, starti...
5:05:1...	MsMpEng.exe	3768	TCP Send	DESKTOP-SNEVEAD:64722 -> 57.155....	SUCCESS	Length: 408, starti...
5:05:1...	MsMpEng.exe	3768	TCP Send	DESKTOP-SNEVEAD:64722 -> 57.155....	SUCCESS	Length: 4392, starti...
5:05:1...	MsMpEng.exe	3768	TCP Send	DESKTOP-SNEVEAD:64722 -> 57.155....	SUCCESS	Leng
5:05:1...	MsMpEng.exe	3768	TCP Send	DESKTOP-SNEVEAD:64722 -> 57.155....	SUCCESS	Leng
5:05:5...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64723 -> tzdelb-b...	SUCCESS	Leng
5:05:5...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64723 -> tzdelb-b...	SUCCESS	Leng
5:05:5...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64724 -> a104-9...	SUCCESS	Length: 2179, starti...
5:05:5...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64724 -> a104-9...	SUCCESS	Length: 80, startim...
5:05:5...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64724 -> a104-9...	SUCCESS	Length: 525, starti...
5:06:0...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64725 -> ec2-34-...	SUCCESS	Length: 1906, starti...
5:06:0...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64725 -> ec2-34-...	SUCCESS	Length: 64, startim...
5:06:0...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64725 -> ec2-34-...	SUCCESS	Length: 92, startim...
5:06:0...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64725 -> ec2-34-...	SUCCESS	Length: 442, starti...
5:06:0...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64725 -> ec2-34-...	SUCCESS	Length: 31, startim...
5:06:0...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64725 -> ec2-34-...	SUCCESS	Length: 124, starti...
5:06:0...	chrome.exe	4304	TCP Send	DESKTOP-SNEVEAD:64725 -> ec2-34-...	SUCCESS	Length: 33, startim...

Length: 158
startime: 2115372
endtime: 2115381
seqnum: 0
connid: 0

UDP

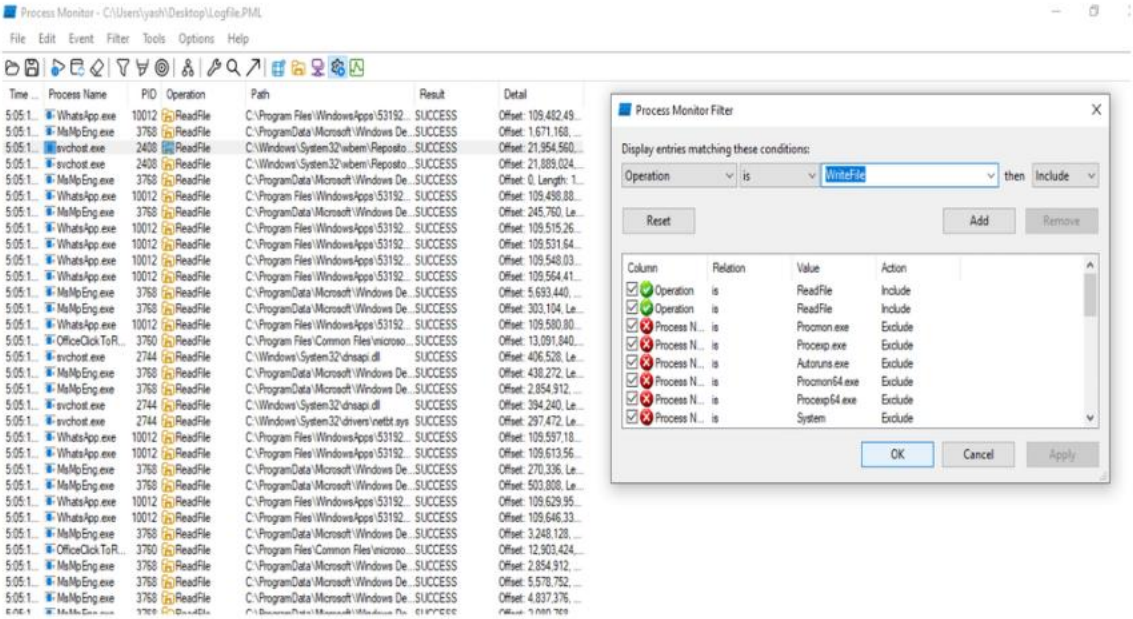
Process Monitor - C:\Users\yash\Desktop\Logfile.PML

File Edit Event Filter Tools Options Help

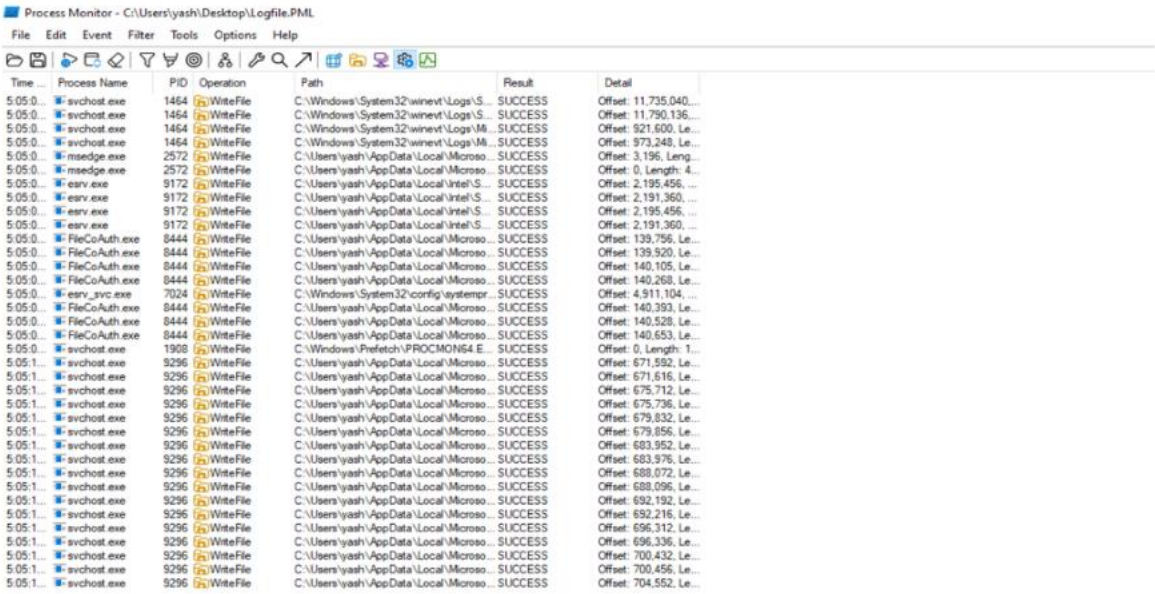
Time ...	Process Name	PID	Operation	Path	Result	Detail
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1094, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:55749 -> bkk02s...	SUCCESS	Length: 1243, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:55749 -> bkk02s...	SUCCESS	Length: 1250, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:55749 -> bkk02s...	SUCCESS	Length: 973, seqn...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 849, seqn...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:57341 -> maa05s...	SUCCESS	Length: 40, seqnu...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1022, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:52845 -> 142.25...	SUCCESS	Length: 48, seqnu...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:55749 -> bkk02s...	SUCCESS	Length: 34, seqnu...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:64502 -> 192.16...	SUCCESS	Length: 52, seqnu...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:56112 -> 192.16...	SUCCESS	Length: 52, seqnu...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 861, seqn...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 861, seqn...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:52845 -> 142.25...	SUCCESS	Length: 48, seqnu...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 876, seqn...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 876, seqn...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 578, seqn...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:52845 -> 142.25...	SUCCESS	Length: 48, seqnu...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1186, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1186, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1186, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1186, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1187, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1187, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1187, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1187, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1187, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1187, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1187, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 590, seqn...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:52845 -> 142.25...	SUCCESS	Length: 65, seqnu...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:55749 -> bkk02s...	SUCCESS	Length: 41, seqnu...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1188, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 1189, seq...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 523, seqn...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:52845 -> 142.25...	SUCCESS	Length: 48, seqnu...
5:05:5...	chrome.exe	4304	UDP Send	DESKTOP-SNEVEAD:61990 -> 142.25...	SUCCESS	Length: 518, seqn...

d. Monitor the Hard disk

Read file



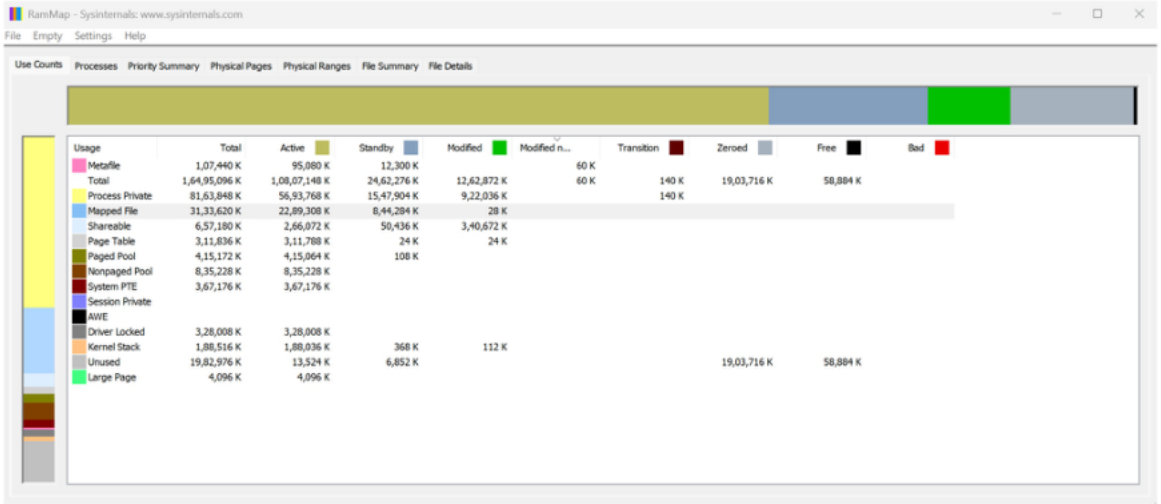
Write file



e. Monitor Virtual Memory

f. Monitor Cache Memory

1. Look for the entries related to cache memory:
    - **Standby**
    - **Modified**
    - **Mapped File**



Usage	Total	Active	Standby	Modified	Modified n...	Transition	Zeroed	Free	Bad
Metafile	1,07,440 K	95,080 K	12,300 K		60 K				
Total	1,64,95,096 K	1,08,07,148 K	24,62,276 K	12,62,872 K	60 K	140 K	19,03,716 K	58,884 K	
Process Private	81,63,848 K	56,93,768 K	15,47,904 K	9,22,036 K		140 K			
Mapped File	31,33,620 K	22,89,308 K	8,44,284 K	28 K					
Shareable	6,57,180 K	2,66,072 K	50,436 K	3,40,672 K					
Page Table	3,11,836 K	3,11,788 K	24 K	24 K					
Paged Pool	4,15,172 K	4,15,064 K	108 K						
Nonpaged Pool	8,35,228 K	8,35,228 K							
System PTE	3,67,176 K	3,67,176 K							
Session Private									
AWE									
Driver Locked	3,28,008 K	3,28,008 K							
Kernel Stack	1,88,516 K	1,88,036 K	368 K	112 K					
Unused	19,82,976 K	13,524 K	6,852 K				19,03,716 K	58,884 K	
Large Page	4,096 K	4,096 K							

2. Now click on the **File Summary** tab.
3. Check which files are stored in **cache memory** — usually **DLLs, EXE files, browser files, system files** etc.
4. Sort by **Total** or **Active** columns to identify which files occupy most cache memory.

RamMap - Sysinternals: www.sysinternals.com

File Empty Settings Help

Use Counts Processes Priority Summary Physical Pages Physical Ranges File Summary File Details

Path	Total	Standby	Modified	Modified No-Write	Active
C:\program files\google\chrome\application\142.0.7444.163\chrome.dll	1,50,552 K	16,904 K			1,33,648 K
C:\program files\common files\adobe\creative cloud libraries\libs\node.exe	1,00,860 K	90,260 K			10,600 K
C:\program files (x86)\microsoft\edgewebview\application\142.0.3595.90\msedge.dll	90,876 K	14,332 K			76,544 K
C:\users\abhishek\appdata\local\programs\perplexity\perplexity.exe	68,544 K	14,132 K			54,412 K
C:\program files\quick heal\quick heal total security\quarantine\bdsquar\ngavdb.db	51,224 K	50,508 K			716 K
C:\windows\system32\driverstore\filerepository\igd_dch.inf_amd64_a582115f5de863e1\igdkmdn64.sys	49,396 K	348 K			49,048 K
C:\users\abhishek\appdata\local\discord\app-1.0.9215\discord.exe	41,748 K	3,852 K			37,896 K
C:\program files\common files\adobe\adobe desktop common\cef\libcef.dll	39,820 K	4,596 K			35,224 K
C:\users\abhishek\appdata\local\google\chrome\user data\default\history	27,908 K	11,256 K			16,652 K
C:\windows\system32\driverstore\filerepository\nvlti.inf_amd64_071505319ec619da\nvlddmkm.sys	26,692 K	8,452 K			18,240 K
C:\program files\quick heal\quick heal total security\ft\fcmcomm.exe	24,500 K	2,124 K			22,376 K
C:\windows\system32\config\software	20,228 K	2,344 K			17,884 K
C:\windows\system32\driverstore\filerepository\igd_dch.inf_amd64_a582115f5de863e1\igc64.dll	13,124 K	1,140 K			11,984 K
C:\windows\system32\windows.ui.xaml.dll	12,288 K	2,068 K			10,220 K
C:\program files\windowsapps\microsoft.windowsstore_22510.1401.2.0_x64__8wekyb3d8bbwe\winstore....	10,736 K	24 K			10,712 K
C:\program files (x86)\adobe\adobe sync\coresync\coresync.exe	10,332 K	68 K			10,264 K
C:\program files\adobe\adobe creative cloud experience\libs\node.exe	10,244 K	32 K			10,212 K
C:\windows\serviceprofiles\localservice\appdata\local\fontcache\~fontcache-fontface.dat	10,240 K	24 K			10,216 K
C:\windows\assembly\nativeimages_v4.0.30319_64\mscorlib\afb0bf793a68e8bf3048cfd9bf9d4a69\mscorL...	10,040 K	1,224 K			8,816 K
C:\windows\systemapps\microsoftwindows.client.cbs_cw5n1h2txyewy\osclient.net.dll	9,260 K				9,260 K
C:\windows\systemapps\microsoft.windowsapprruntime.cbs_8wekyb3d8bbwe\microsoft.ui.xaml.dll	8,640 K	196 K			8,444 K
C:\windows\serviceprofiles\localservice\appdata\local\fontcache\fontcache-obsolete-171.dat	8,628 K	8,592 K	8 K		28 K
C:\windows\system32\driverstore\filerepository\nvlti.inf_amd64_071505319ec619da\nvgpucomp64.dll	7,904 K	5,704 K			2,200 K
C:\windows\system32\mshtml.dll	7,800 K	7,800 K			
C:\windows\system32\taskmgr.exe	7,600 K	5,788 K			1,812 K